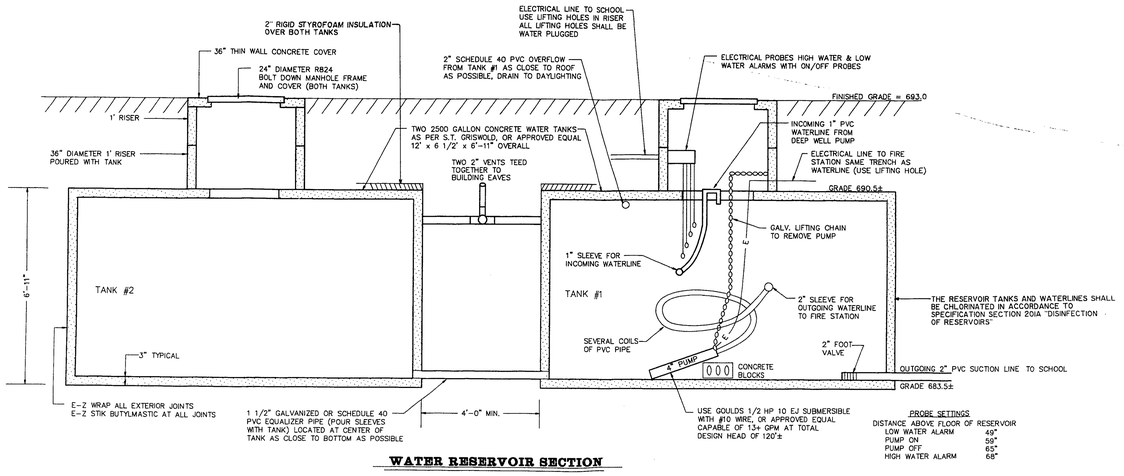
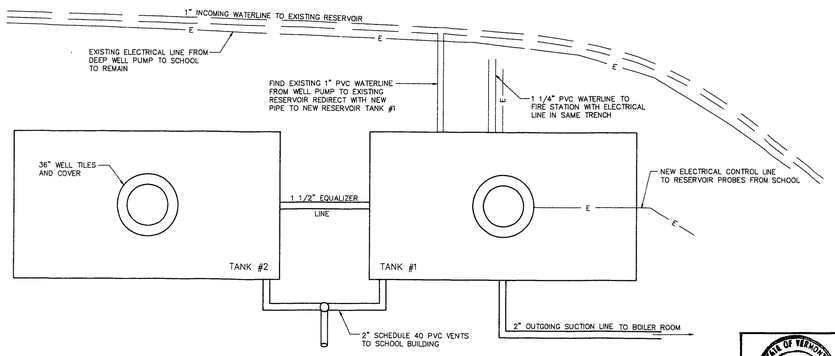
WW-5-0204-1



WATER RESERVOIR SECTION



WATER RESERVOIR LAYOUT SCHEMATIC

REVISED 5-14-91

Dept. of Environmental Conservation
Approved: [signature]
Permit #: WW-5-0204-1
Date: 6/17/91

RECEIVED
MAY 23 1991
DIVISION OF [illegible]

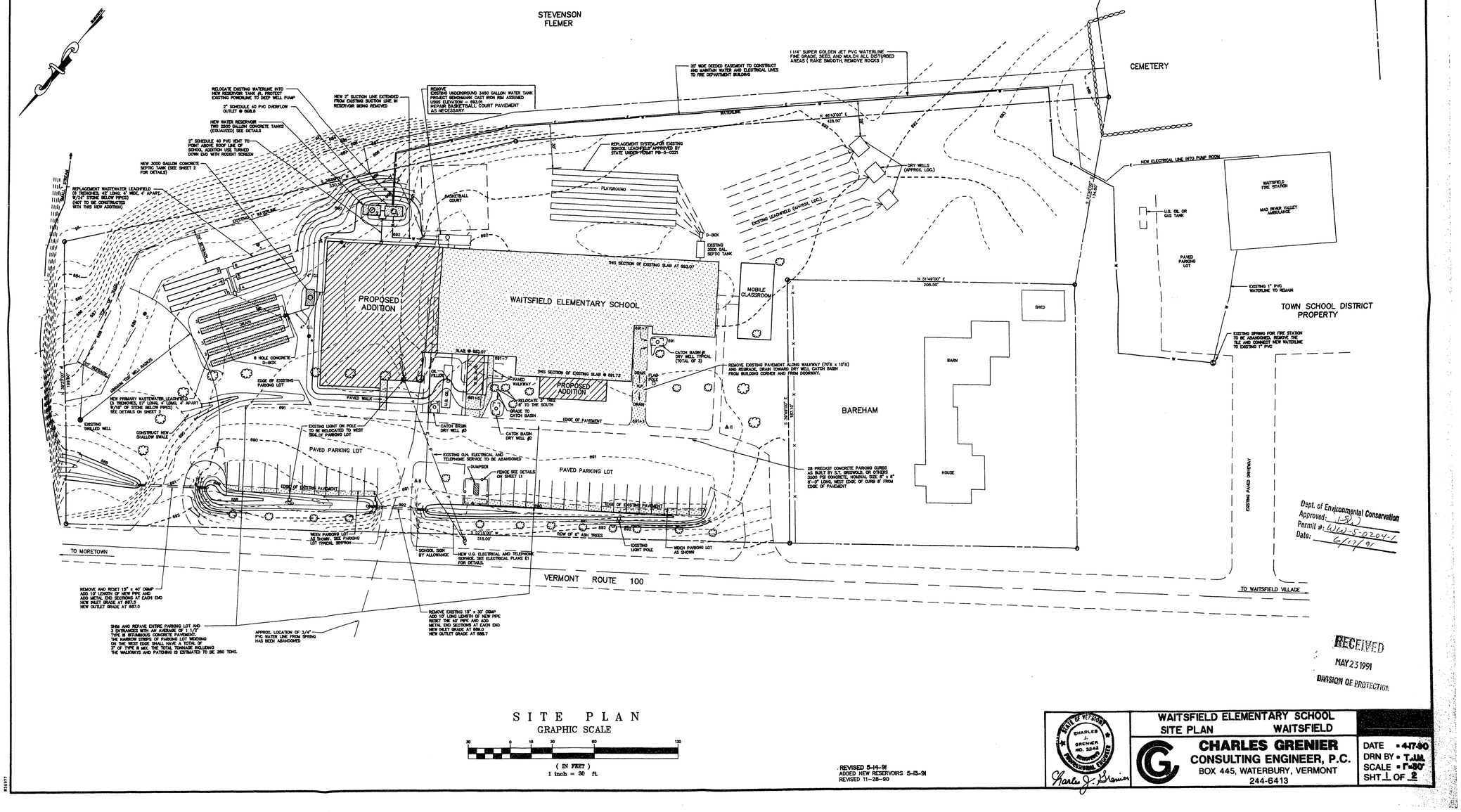
STATE OF VERMONT
CHARLES J. GRENIER NO. 3245
REGISTERED PROFESSIONAL ENGINEER

Charles J. Grenier

WAITSFIELD ELEMENTARY SCHOOL
WATER SYSTEM DETAILS

CHARLES GRENIER
CONSULTING ENGINEER
BOX 445, WATERBURY, VERMONT
244-6413

DATE - 5-13-91
DRN BY - T.J.M.
SCALE - A.N.
SHT 1A OF 2



SITE PLAN

GRAPHIC SCALE

30 0 15 30 60 120

(IN FEET)

1 inch = 30 ft.

REVISED 5-14-91
ADDED NEW RESERVOIRS 5-13-91
REVISED 11-28-90

STATE OF VERMONT — CHARLES J. GRENIER NO. 5248 — REGISTERED PROFESSIONAL ENGINEER

Charles J. Grenier

WAITSFIELD ELEMENTARY SCHOOL
SITE PLAN WAITSFIELD

CHARLES GRENIER
CONSULTING ENGINEER, P.C.
BOX 445, WATERBURY, VERMONT
244-6413

DATE = 4-17-90
DRN BY = T.J.M.
SCALE = 1"=30'
SHT 1 OF 2

R15577

WW-5-0204-1

Also See: WW-5 0298

PROJECT DOCKET SHEET

ID# WIU-S-0204-1

DATE	REVIEW ACTION
5/23/91	Info rcvd
5/27/91	Logged in, 1 copy to PS

Project ID #: WW-5-0204-1 C = CO Applicant O = Owner S = Seller

APPLICANT

Name: Waitsfield School District

Address:

City, State, Zip: Waistfield, VT 05673

CROSS REFERENCE

Waitsfield Fire Department

WW-5-0298

Waitsfield, VT 05673

Town of Devt: WAITSFIELD

Road/River: RT 100

Devt Name:

Description: replace water reservoir

Fee Rec'd: $

DATES

Rec'd at Dist: ~~XXXXXX~~ 5/23/91
Logged/Rec Ltr: 5/27/91
Initial Review: 6/17/91
Comp. Response: 6/17/91
Final Action: 6/17/91
Type of Final Action: P
(See list of Final Action Codes Below)

Consulting Engineer: Charles Grenier Box 445 Waterbury 05676

Hydrogeologist:

Reg. Review Eng: WERNECKE

Action Requested:

FINAL ACTION CODES

P = Permit
D = Denial
C = Certification of Compliance
N = Cert. of Noncompliance
W = Withdrawn

TO BE COMPLETED AFTER FINAL ACTION

Minor Project (Y/N) N
Type of Water Supply (ON/OFF/MUN/COM/NO):
Type of Sewage Disposal (ON/OFF/MUN/COM/NO):
Date Permit Recorded:

Number of Lots Approved: 0
Number of Mounds: 0
Gals. Sewage Generated:
GPD Water Supply

FOR SUBDIVISIONS ONLY:
List deferral number if this permit removes a deferral:

FOR DEFERRALS ONLY:
Is parcel retained or conveyed (R/C):
Is parcel contiguous or noncontiguous to other lands of buyer (C/N):

FOR STREAM ALTERATION PERMITS ONLY:
PURPOSE (SM = strm Stab/Mod PB = Private Bridge UC = Utility Crossing GR = Gravel Removal SR = Stream Reloc/Realign OT = Other):

FOR HIGHWAY DEPARTMENT PROJECT ONLY:
PURPOSE (BR = Bridge RD = Roadway CM = Channel Maint CT = Culvert OT = Other):

State of Vermont

Department of Fish and Wildlife
Department of Forests, Parks and Recreation
Department of Environmental Conservation
State Geologist
Natural Resources Conservation Council

AGENCY OF NATURAL RESOURCES
Department of Environmental Conservation
Barre Regional Office
324 North Main Street
Barre, VT 05641

Telephone 479-3621

January 20, 1992

Waitsfield School District
Waitsfield, VT 05673

RE: WW-5-0204-1, Waitsfield Elementary School, water system, Waitsfield

Gentlemen:

This letter is a follow up to the information submitted for the above-referenced permit.

Please be advised that conditions 2.2 and 2.3 of the above-referenced permit have been satisfied.

If you have any questions please contact this office.

Sincerely,

Don Wernecke
Assistant Regional Engineer

CC Charles Grenier
Richard Moser
Waitsfield Fire Dept.

CHARLES GRENIER
CONSULTING ENGINEER PC
BOX 445 244-6413
WATERBURY, VT. 05676

September 19, 1991

RECEIVED
SEP 25 1991
DEPARTMENT OF PROTECTION

Don Wernecke, Assistant Regional Engineer
District #5 Environmental Office
324 North Main Street
Barre, Vermont 05641

RE: Waitsfield Elementary School and Waitsfield Fire Department Water System, Permit WW5-0204-1 and WW5-0298

Dear Don:

As required by Section 2 of the above mentioned permits, our office was retained to inspect the installation of the water system and to report changes. The reservoir tanks were installed and disinfected according to the specifications and the pump installed in the reservoir to serve the Fire Station. The attached water sample was taken revealing no bacteria. The Fire Department and the School Board agreed to install a new pressure tank and control system in a small utility room at the south east corner of the new school addition. This new utility room is near the location of the new reservoirs. We have prepared the attached System Schematic. The pump in the reservoir is controlled by a pressure switch in the new utility room. Water is pumped into a new WX203 extrol tank and then fed through the building wall to the Fire Station. The existing extrol tank and centrifigal pump in the Fire Station, which was part of the old water system, has been removed.

It is our opinion that all conditions of the State permits relating to water and sewer systems have been met. If any questions arise, please do not hesitate to contact us.

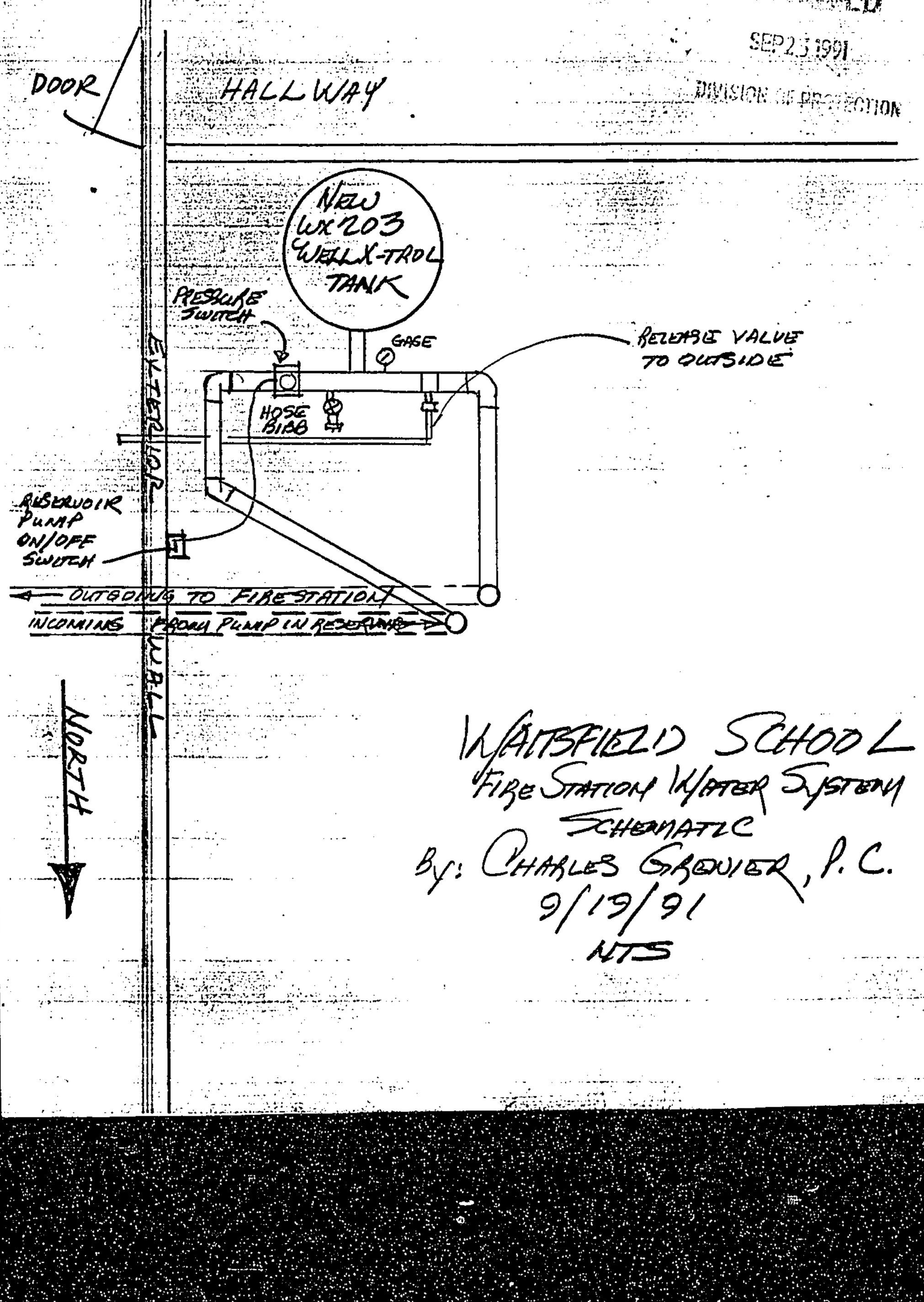
Respectfully,

Charles J. Grenier
Consulting Engineer, P.C.

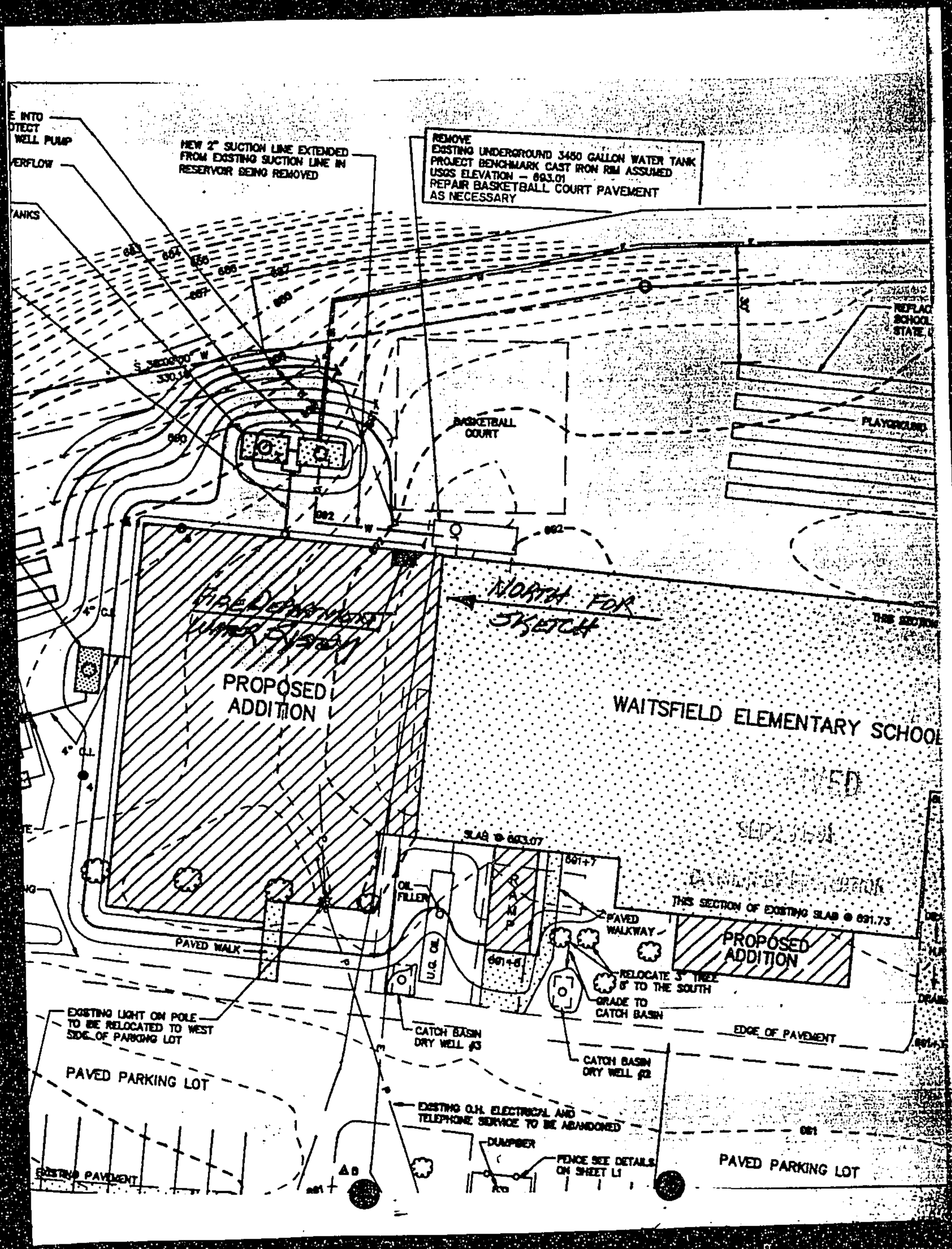
CJG:lar

cc. Waitsfield Elementary
Waitsfield Fire Department
Black River Design
Scotts Construction

RECEIVED
SEP 23 1991
DIVISION OF PROTECTION



WAITSFIELD SCHOOL
FIRE STATION WATER SYSTEM
SCHEMATIC
By: CHARLES GROVER, P.E.
9/19/91
NTS


NEW 2" SUCTION LINE EXTENDED FROM EXISTING SUCTION LINE IN RESERVOIR BEING REMOVED
REMOVE
EXISTING UNDERGROUND 3450 GALLON WATER TANK
PROJECT BENCHMARK CAST IRON RIM ASSUMED
USGS ELEVATION – 693.01
REPAIR BASKETBALL COURT PAVEMENT AS NECESSARY
BASKETBALL COURT
PLAYGROUND
NORTH FOR SKETCH
PROPOSED ADDITION
WAITSFIELD ELEMENTARY SCHOOL
SLAB @ 693.07
THIS SECTION OF EXISTING SLAB @ 691.73
PROPOSED ADDITION
PAVED WALKWAY
RELOCATE 3" TREE 8' TO THE SOUTH
GRADE TO CATCH BASIN
OIL FILLER
U.G. OIL
PAVED WALK
EXISTING LIGHT ON POLE TO BE RELOCATED TO WEST SIDE OF PARKING LOT
CATCH BASIN DRY WELL #3
CATCH BASIN DRY WELL #2
EDGE OF PAVEMENT
PAVED PARKING LOT
EXISTING O.H. ELECTRICAL AND TELEPHONE SERVICE TO BE ABANDONED
DUMPSTER
FENCE SEE DETAILS ON SHEET L1
PAVED PARKING LOT
EXISTING PAVEMENT

POOR COPY
RECEIVED FOR
MICROFILMING

RECEIVED
SEP 16 1991
SCOTT'S CONSTRUCTION INC

SPECTRUM
Environmental Research
Laboratories, Inc.
P.O. Box 122
Montpelier, Vt. 05602
(802) 223-7088

SEP 2 1991
INSPECTION

Report To: Viens P & H
PO Box 1976
Waitsfield VT 05673

CC: Scotts Construction
Newport Vt

Sampled By: K. Viens P&H

Date Received: September 11,1991

Date of Report: September 13,1991

These samples were analyzed according to "Standard Methods for Examination of Water and Wastewater", Latest Edition, APHA, AWWA, and WPCF.

Sample #	Client Identification	Tests Performed	Results
6293	Waitsfield School Waitsfield VT 05673 **Bathroom sink old section**	Total Coliform	0 per 100 mls *

* This sample **does** meet State Health Department microbiological standards for drinking water.

Analyst Charlotte D. Sherman

State of Vermont

AGENCY OF NATURAL RESOURCES

WATER SUPPLY AND WASTEWATER DISPOSAL PERMIT

CASE No: WW-5-0204-1 (REVISED)
APPLICANT: Waitsfield School District
ADDRESS: Waitsfield, VT 05673

LAWS/REGULATIONS INVOLVED
10 V.S.A. Chapter 61, Water Supply and Wastewater Disposal
Environmental Protection Rules
Chapter 4, Public Buildings
Chapter 8, Water Supply
Appendix A, Design Guidelines

This project, consisting of an after school child care program for a maximum of 26 children and 2 staff, located at the Waitsfield Elementary School, Route 100, Waitsfield, Vermont, is hereby approved under the requirements of the regulations named above, subject to the following conditions:

1. GENERAL CONDITIONS

1.1. The project must be completed as shown on the plans listed in Water Supply and Wastewater Disposal permit number WW-5-0204-1 dated 6/17/91, and which have been stamped "APPROVED" by the Division of Protection. No alteration of these plans shall be allowed except where written application has been made to the Agency of Environmental Conservation and approval obtained.

1.2. The approved maximum student population at the school is 300. The maximum approved enrollment in the after school child care program is 26 children and 2 staff. Prior to any increase in these enrollments, approval must be obtained from the Division of Protection.

1.3. This authorization does not relieve you, as applicant, from obtaining all approvals and permits as may be required from the Department of Labor and Industry (phone 828-2106), the Vermont Department of Social and Rehabilitavie Services, and local officials PRIOR to construction.

2. WATER CONDITIONS

2.1. All water conditions set forth in Water Supply and Wastewater Disposal Permit WW-5-0204-1, dated 6/17/91, shall remain in effect.

3. SEWAGE CONDITIONS

3.1. All sewage conditions set forth in Water Supply and Wastewater Disposal Permit number WW-5-0204-1, dated 6/17/91, shall remain in effect.

Reginald A. LaRosa, Acting Commissioner
Department of Environmental Conservation

By _____ 8/29/91
Don Wernecke, Regional Engineer

CC Charles Grenier
VT Dept. of Labor & Industry
VT Dept. of SRS
Town Planning Commission
Don Robisky

Rev. 10/1/90

AGENCY OF NATURAL RESOURCES
DEPARTMENT OF ENVIRONMENTAL CONSERVATION
PERMITS, COMPLIANCE & PROTECTION DIVISION

FOR OFFICE USE ONLY
Permit #_____
Date Application Received_____

WATER SUPPLY AND WASTEWATER DISPOSAL
(Schedule "C" to Master Act 250 Land Use Permit Application)

THERE IS TO BE NO SITE WORK OR CONSTRUCTION COMMENCED ON THIS PROJECT WITHOUT WRITTEN APPROVAL FROM THE AGENCY OF NATURAL RESOURCES.

RECEIVED
AUG 27 1991
DIVISION OF PROTECTION

LANDOWNER:
Name: TOWN OF WAITSFIELD SCHOOL DISTRICT (Waitsfield Elementary School)
Address: RT 100 WAITSFIELD, VT
(Mailing) RR1 Box 397 Waitsfield VT
Tele. No.: 496-3643 / 496-5429

CO-APPLICANT:
Name:
Address:
(Mailing)
Tele. No.:

Business Name (if other than above): Waitsfield Recreation Program for School Age Children

CONSULTANT: (architect, engineer, plumber) ~~see file~~
Name: None Tele. No.

Address:
(mailing) (Street) (Town) (State) (Zip)

A. PROJECT LOCATION: (Attach Map)

Town: Waitsfield Road/Highway: Rt 100

B. When was the parcel created? 1960 I

C. List any prior Environmental or Act 250 permits issued for A above: WW-5-0204 remands WW-5-0298

D. Describe the project (be specific: see item #1 on back) (Attach 2 sets of plans)
To provide after school childcare for school aged children (26 children; 2 staff). Monday - Friday from 2:45 - 5:30pm. Enrollment will not be limited to existing school population.

E. TYPE OF WATER SOURCE: (Municipal, offsite, community, onsite) onsite
(Attach approval - see #8 on back)

F. Does this water supply serve 2 or more buildings? no

G. TYPE OF SEWAGE DISPOSAL: (municipal, offsite, community, onsite) onsite
(if municipal, attach approval - see #8 on back)

H. Number of gallons of sewage generated: Existing Flows: see file 4500 approved WW-5-0204 New Flows:

I. APPLICATION FEE: ($0.22 per gallon of design flow of project - minimum fee of $100.00 per application.) Make check payable to STATE OF VERMONT.
AMOUNT ENCLOSED $ Exempt - municipality

27 August 1991 — Carol A. Burkland BOARD CLERK
DATE — SIGNATURE OF LANDOWNER

DATE — SIGNATURE OF CO-APPLICANT

PROJECT INFORMATION SHEET
Water Supply and Wastewater Disposal

( ) 1. Description of Project (on application form):
Should include number of seats if a restaurant, or number of living units if an apartment complex, number of square feet, number of employees or similar figures, type of water supply, type of sewage disposal, description of process wastes. If project is a change in use, describe current use.

( ) 2. Location on Project on USGS map or Town Highway Map.

( ) 3. Plot Plan (submit 2 copies) by a professional engineer registered in the State of Vermont showing:

( ) building location and lot dimensions;
( ) existing and proposed surface waters and drainage systems;
( ) location of existing and proposed sewage system and/or sewer lines;
( ) location of existing and proposed water system and water supply lines;
( ) five foot contours in the sewage disposal area;
( ) location of all soil testing and bedrock outcrops;
( ) use of adjoining lands and location of their water and sewage systems.

( ) 4. Floor plans indicating use of each interior space, sanitary facilities and plumbing fixture schedule. Submit 2 copies prepared by a registered architect, engineer or other qualified consultant.

( ) 5. Soil data and sewage system design information by a professional engineer registered in the State of Vermont (submit 2 copies):

( ) test borings and soils description, and percolation tests in the area of the proposed sewage system;
( ) design of the proposed sewage system based on design flows and the soils information;
( ) design of existing sewage system and statement as to the capability and condition of the system;
( ) building sewer and later plan, profile, and sections, construction, material and testing specifications.

( ) 6. Water supply system designed by a registered professional engineer (submit 2 copies):

( ) approval by Health Department, Water Supply Section required. Contact the Health Department at 862-7220;
( ) water source yield test;
( ) a water supply system design;
( ) report of ( ) chemical and/or ( ) bacteriological testing of water supply;
( ) design of existing water supply system and statement as to the capability and condition of the system.

( ) 7. All plans to be dated and signed by designer.

( ) 8. Municipal approval for connection of ( ) water supply ( ) wastewater disposal (Submit letter from municipality approving connections)

( ) 9. Application Fee: $0.22 per gallon of daily design flow of project - minimum fee of $100.00 per application. Make check payable to the STATE OF VERMONT.

August 1991

RECEIVED

AUG 27 1991

DIVISION OF PROTECTION

Agency of Natural Resources

Beginning in September we plan to offer an after school child care program at the Waitsfield Elementary School. The program will be from 2:45 until 5:30 Monday through Friday.

In the Spring of 1991, a parent questionaire/survey demonstrated a strong interest and need for after school childcare in the Waitsfield population.

The program will offer a variety of activities including sports, games, crafts, and quiet activities. No meals will be served, though a simple snack will be offered early in the day.

We are being licensed for 26 children, though we anticipate the daily attendance to be approximately fifteen.

We have applied for tax exempt, non profit status.

Carol Rauch Bellage
Board Member

State of Vermont

AGENCY OF NATURAL RESOURCES
WATER SUPPLY AND WASTEWATER DISPOSAL PERMIT

CASE No: WW-5-0204-1
APPLICANT: Waitsfield School District
ADDRESS: Waitsfield, VT 05673

LAWS/REGULATIONS INVOLVED
10 V.S.A. Chapter 61, Water Supply and Wastewater Disposal Environmental Protection Rules
Chapter 4, Public Buildings
Chapter 8, Water Supply
Appendix A, Design Guidelines

This project, consisting of replacing a water supply pump station serving the Waitsfield Elementary School, located on Route 100, Waitsfield, Vermont, is hereby approved under the requirements of the regulations named above, subject to the following conditions:

1. GENERAL CONDITIONS

1.1. The project must be completed as shown on the plans prepared by Charles Grenier listed as follows:
Site Plan, Dated 4/17/90, last revised 5/14/91,
Water System Details, Dated 5/13/91, last revised 5/14/91,
and which have been stamped "APPROVED" by the Division of Protection. No alteration of these plans shall be allowed except where written application has been made to the Agency of Environmental Conservation and approval obtained.

1.2. A copy of the approved plans and the Water Supply and Wastewater Disposal Permit shall remain on the project during all phases of construction and, upon request, shall be made available for inspection by State or Local personnel.

1.3. No alterations to the existing building other than those indicated on the approved plan, which would change or affect the interior waste plumbing, water supply, or wastewater disposal shall be allowed without prior review and approval from the Agency of Environmental Conservation.

1.4. By acceptance of this permit the permittee agrees to allow representatives of the State of Vermont access to the property covered by the permit, at reasonable times, for the purpose of ascertaining compliance with Vermont environmental and health statutes and regulations and with the permit.

1.5. This permit shall in no way relieve you of the obligations of Title 10, Chapter 48, Subchapter 4, for the protection of groundwater.

2. WATER CONDITIONS

2.1. This project is approved with an on-site water supply from an existing drilled well provided that the well is located as shown on the plans and no closer than 100 feet to any subsurface wastewater disposal system, 50 feet from any septic tanks, and 50 feet from any sewer lines.

2.2. The installation of the water supply system shall be inspected by a Vermont registered Professional Engineer who shall report in writing, prior to occupancy, that the work has been completed in accordance with the approved plans.

2.3. Prior to occupancy of the Elementary School, the water distribution system shall be adequately chlorinated, thoroughly flushed and subsequently tested for bacteriological content. The system shall only be put into use after favorable water test results are obtained. A copy of the results must be submitted to this office.

(CONTINUED)

3. SEWAGE CONDITIONS

3.1. This project is approved with an existing subsurface wastewater disposal system. Should this system ever malfunction, the permittee must construct the replacement sewage disposal system in accordance with the requirements of WW-5-0204 dated 5/24/90, the approved plans and the conditions of this permit. The permittee shall report any failure of the existing disposal system to the Protection Division within 48 hours.

Reginald A. LaRosa, Acting Commissioner
Department of Environmental Conservation

By _____ 6/17/91
Don Wernecke, Assistant Regional Engineer

CC Charles Grenier
Town Planning Commission
Don Robisky

State of Vermont

Department of Fish and Wildlife
Department of Forests, Parks and Recreation
Department of Environmental Conservation
State Geologist
Natural Resources Conservation Council

AGENCY OF NATURAL RESOURCES
Department of Environmental Conservation
Barre Regional Office
324 North Main Street
Barre, VT 05641

Telephone 479-3621

May 27, 1991

Waitsfield School District
Waitsfield, VT 05673

Dear Applicant:

RE: WW-5-0204-1 , replace water reservoir, WAITSFIELD

We received your completed application for the above referenced project on May 23, 1991, including a fee of $0 paid by check number . This application falls under the Water/Wastewater: Less than 500 GPD program; and under the Performance Standards for this program area, we have a maximum of 30 days of "in-house" time for our review of your application.

Based upon the present workload, we anticipate review of your project within the next one to two weeks.

The checklist on the back of this letter indicates other State agencies you should contact regarding any approvals you may need under their programs. The information contained in your application has been made available to an Act 250 District Coordinator for a determination of whether your project will also require an Act 250 permit. If you have questions relating to Act 250 jurisdiction, a District Coordinator may be reached at 479-3621 for Washington and Lamoille County projects and for projects in the towns of Orange, Washington, and Williamstown. For other Orange County projects call the District Coordinator at 886-2215.

If you have not already done so, you should check with town officials regarding any local requirements you may need to meet.

For the Division of Protection

Alida J. Conti
Administrative Secretary

cc: Act 250 District Coordinator
WAITSFIELD Planning Commission
Charles Grenier
Waitsfield Fire Department

SEE REVERSE FOR OTHER STATE AGENCIES THAT MAY REQUIRE APPROVAL OF THIS PROJECT

Rev. 10/1/90

AGENCY OF NATURAL RESOURCES
DEPARTMENT OF ENVIRONMENTAL CONSERVATION
PERMITS, COMPLIANCE & PROTECTION DIVISION

FOR OFFICE USE ONLY
Permit # WW-5-0204-1
Date Application Received RECEIVED
MAY 23 1991
DIVISION OF PROTECTION

WATER SUPPLY AND WASTEWATER DISPOSAL
(Schedule "C" to Master Act 250 Land Use Permit Application)

THERE IS TO BE NO SITE WORK OR CONSTRUCTION COMMENCED ON THIS PROJECT WITHOUT WRITTEN APPROVAL FROM THE AGENCY OF NATURAL RESOURCES.

LANDOWNER:
Name: Waitsfield School District .

Address: Waitsfield, Vermont 05673

(Mailing) _____

Tele. No.: _____

CO-APPLICANT:
Name: Waitsfield Fire Department

Address: Waitsfield, Vermont 05673

(Mailing) _____

Tele. No.: _____

Business Name (if other than above): _____

CONSULTANT: (architect, engineer, plumber)
Name: Charles J. Grenier, Consulting Engineer, P.C. Tele. No. 244-6413

Address: P.O. Box 445, Waterbury, Vermont 05676
(mailing) (Street) (Town) (State) (Zip)

A. **PROJECT LOCATION:** (Attach Map)

Town: Waitsfield Road/Highway: Route 100

B. When was the parcel created? 1960±

C. **List any prior Environmental or Act 250 permits issued for A above:** WW5-0204

D. Describe the project (be specific: see item #1 on back) **(Attach 2 sets of plans)**
Replace existing reservoir (3450 gallons) with new 5000 gallon concrete reservoir (allocating 4500 gallons for School and 500 gallons for Fire Department). The Fire Department will derive its water supply from the School well. A pump to supply the Fire Department will be installed in the new reservoir.

E. **TYPE OF WATER SOURCE:** (Municipal, offsite, community, onsite) on-site
(Attach approval - see #8 on back)

F. Does this water supply serve 2 or more buildings? Yes

G. **TYPE OF SEWAGE DISPOSAL:** (municipal, offsite, community, onsite) on-site
(if municipal, attach approval - see #8 on back)

H. Number of gallons of sewage generated: Existing Flows: 4500± New Flows: 500±

I. **APPLICATION FEE:** ($0.22 per gallon of design flow of project - minimum fee of $100.00 per application.) Make check payable to STATE OF VERMONT.
AMOUNT ENCLOSED $ Not applicable

5/15/91 _____
DATE **SIGNATURE OF LANDOWNER**

5/15/91 _____
DATE **SIGNATURE OF CO-APPLICANT**

PROJECT INFORMATION SHEET
Water Supply and Wastewater Disposal

( ) 1. Description of Project (on application form):
Should include number of seats if a restaurant, or number of living units if an apartment complex, number of square feet, number of employees or similar figures, type of water supply, type of sewage disposal, description of process wastes. If project is a change in use, describe current use.

( ) 2. Location on Project on USGS map or Town Highway Map.

( ) 3. Plot Plan (submit 2 copies) by a professional engineer registered in the State of Vermont showing:

- ( ) building location and lot dimensions;
- ( ) existing and proposed surface waters and drainage systems;
- ( ) location of existing and proposed sewage system and/or sewer lines;
- ( ) location of existing and proposed water system and water supply lines;
- ( ) five foot contours in the sewage disposal area;
- ( ) location of all soil testing and bedrock outcrops;
- ( ) use of adjoining lands and location of their water and sewage systems.

( ) 4. Floor plans indicating use of each interior space, sanitary facilities and plumbing fixture schedule. Submit 2 copies prepared by a registered architect, engineer or other qualified consultant.

( ) 5. Soil data and sewage system design information by a professional engineer registered in the State of Vermont (submit 2 copies):

- ( ) test borings and soils description, and percolation tests in the area of the proposed sewage system;
- ( ) design of the proposed sewage system based on design flows and the soils information;
- ( ) design of existing sewage system and statement as to the capability and condition of the system;
- ( ) building sewer and later plan, profile, and sections, construction, material and testing specifications.

( ) 6. Water supply system designed by a registered professional engineer (submit 2 copies):

- ( ) approval by Health Department, Water Supply Section required. Contact the Health Department at 862-7220;
- ( ) water source yield test;
- ( ) a water supply system design;
- ( ) report of ( ) chemical and/or ( ) bacteriological testing of water supply;
- ( ) design of existing water supply system and statement as to the capability and condition of the system.

( ) 7. All plans to be dated and signed by designer.

( ) 8. Municipal approval for connection of ( ) water supply ( ) wastewater disposal (Submit letter from municipality approving connections)

( ) 9. Application Fee: $0.22 per gallon of daily design flow of project - minimum fee of $100.00 per application. Make check payable to the STATE OF VERMONT.

CHARLES GRENIER
CONSULTING ENGINEER PC
BOX 445 244-6413
WATERBURY, VT. 05676

May 21, 1991

RECEIVED
MAY 23 1991
DEPARTMENT OF PROTECTION

Don Wernecke, Assistant Regional Engineer
District #5 Environmental Office
324 North Main Street
Barre, Vermont 05641

RE: Waitsfield Elementary School

Dear Don:

If at all possible, could you speed up the plan review of the new reservoirs for the Waitsfield Elementary School. The School Board and the Contractor are in a big rush to order the tanks and install the water reservoirs, septic tanks, and leachfields at the same time. All site work must be finished, graded, and grass growing prior to the start of school in September. Your prompt attention to this matter would be greatly appreciated.

Respectfully,

Charles J. Grenier
Consulting Engineer, P.C.

CJG:lar

cc. Carol Hosford
Delbert Palmer

CHARLES GRENIER
CONSULTING ENGINEER PC
BOX 445 244-6413
WATERBURY, VT. 05676

May 14, 1991

RECEIVED
MAY 23 1991
DIVISION OF PROTECTION

Don Wernecke, Assistant Regional Engineer
District #5 Environmental Office
324 North Main Street
Barre, Vermont 05641

RE: Waitsfield School and Fire Department Water System

Dear Don:

As you are aware, the Waitsfield Fire Department has an inadequate supply of water from their existing spring. It has been their desire to obtain water from the existing drilled well which serves the Waitsfield Elementary School. The Fire Department and School Board have met regarding this issue and agreements have been finalized. With this letter we will summarize the engineering drawings and update you as to the conformance of these plans with Water Supply and Wastewater Disposal Permit WW5-0204. As the project relates to the School, we offer the following comments.

1. Condition 1.2 of Permit WW5-0204 limited the existing School water system to a capacity of 3450 gallons per day, or 230 students. The School is proposing to add two new 2500 GPD concrete reservoirs with 4500 gallons of storage being allocated for the School's use and 500 gallons for the Fire Department. With this change, the School now has both water and sewer capacity for up to 300 students.

2. The existing 3450 gallon metal reservoir will be removed.

3. The two reservoirs have been equiped with an overflow and are vented. The existing centrifugal pump in the boiler room will remain in use and will draft water from Tank #1. Tank #1 and #2 are tied together with an equalizer pipe to serve as one reservoir.

4. We have revised the site plan to show the location of the two reservoirs and lines. We have also prepared a separate detail sheet with a layout and cross section of the reservoirs.

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5. In order to maintain a 25' isolation distance from the existing School leachfield, the water line will be constructed within a 20' wide easement on property owned by Stevenson Flemer. The School has received approval from Flemer for this easement. The easement is being drafted by an attorney and it is our understanding that signatures will be forthcoming very shortly.

6. We have attached a copy of a signed Water Supply and Wastewater Disposal permit. The School is the Landowner and the Fire Department the co-applicant. No fee has been included because this is a municipal project.

As this project relates to improvements to the water system for the Waitsfield Fire Station, we offer the following summary.

1. The Fire Department uses the building for monthly meetings. The building is not used for social events such as BINGO. Two times per week 40 Senior Citizens are served meals at the Fire Station. Using 8 gallons per person for food preparation yields 320 gallons of water usage for the meals. The Fire Station has one bathroom with a toilet and lavatory and a small kitchenette area with a double bay kitchen sink and a dishwasher. It is our opinion that allocating a water supply of 500 gallons per day is very conservative and will meet the daily demands of current use at the Fire Department.

2. In the pump room on the back side of the Fire Station there is a small 1/2 horsepower Fairbanks Morse Centrifugal pump which drafts water from the existing spring. This pump operates on a 30/50 pressure cycle and probably draws at a peak of 9± GPM. Our calculations recommend that a new Goulds 10EJ 1/2 horse power submersible pump be placed in reservoir tank #1 to meet the needs of the Fire Department. This pump will produce near 15 GPM at 40 PSI which is adequate to meet the peak demands for the Fire Station.

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It is our opinion that with this long term capital improvement, the water supply for both the Fire Station and the School will meet today's Environmental Standards and will prove to be a great long term benefit to both facilities.

If any questions arise as a result of this information, please do not hesitate to contact us.

Respectfully,

Charles J. Grenier
Consulting Engineer, P.C.

CJG:lar

cc. Carol Hosford
Delbert Palmer

CHARLES [illegible]ENIER
Consulting Engineer
P. O. Box 445
WATERBURY, VERMONT 05676
Phone 244-6413

JOB WATER SYSTEM
WATERFORD SCHOOL / FIRE DEPARTMENT
SHEET NO. 1 OF
CALCULATED BY CJL DATE 5/14/91
CHECKED BY DATE
SCALE

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EXISTING FIRE DEPT. CENT. PUMP SUCTION LIFT 10'
30/50 PSI ½ Hp FAIRBANKS / MORSE
AVE. 9 GPM CAPACITY
HAS PERFORMED SATISFACTORY FOR MANY YEARS

TRY DEEP WELL 4" SUBMERSIBLE / GOULDS ¾ Hp 10EJ
W/ 900' OF 1¼" LINE

@ 12 GPM 2.0'/100' or 18' FRICTION LOSS
8' STATIC
PRESSURE @ 40 PSI 92'
TDH. = 118'

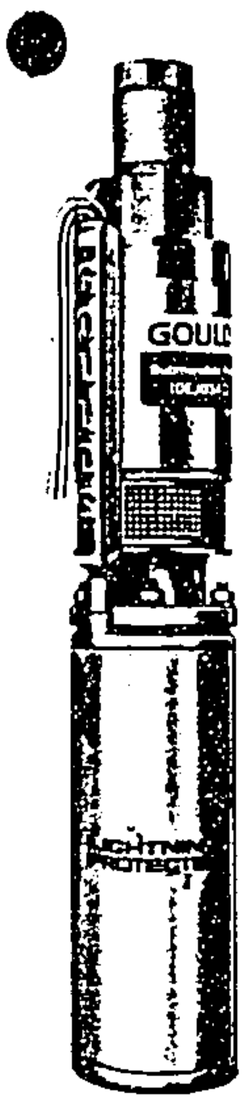
PUMP CURVE SAYS 10.7 gpm

½ Hp 10 EJ SAYS 13.5 gpm

@ 50 psi TDH = 141'
¾ Hp = 8 gpm
½ Hp = 12.4 gpm

SAME PRICE FOR BOTH PUMPS $430.00
SO GO WITH ½ Hp 10 EJ AND
GET MUCH BETTER PERFORMANCE

@ 30 psi TDH = 95'
¾ Hp = 12 gpm
½ Hp = 15 gpm



Goulds

Submersible Pumps

MODEL

10EJ

For 4" and larger wells

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SPECIFICATIONS

Capacities: to 18 GPM
Best efficiency point: 10 GPM
Horsepower: ½-3 HP
Discharge connection: ½-¾ HP — 1" NPT, 1-3 HP — 1¼ NPT
Rotation: Counterclockwise when looking into discharge.
Recommended for 4" wells or larger.

FEATURES

Powered for Continuous Operation: All ratings are within the working limits of the motor as recommended by the motor manufacturer. Pump can be operated continuously without damage to the motor.

Completely Field Serviceable: No special tools needed for repair and no shipping back to factory.

Abrasion-Resistant Construction: Special design water passages permits handling sandy water.

Corrosion-Free Components: Diffusers and impellers made of rugged thermoplastic — resists mineral and algae deposits.

Stainless Steel Bowls: Stack up and rabbet lock together.

Polished Stainless Steel Casing: Durable in the most corrosive waters.

No Gaskets: Interstage leakage controlled by compression of diffuser between bowls.

Cast Bronze Motor Adapters and Discharge Heads: Are on all four inch submersible pumps.

Rugged Check Valve: Positive sealing, cast bronze with cast bronze poppet, monel spring and stainless steel self locking nut, and retainer.

Urethane Discharge Head Bearing: Replaceable bearing allows excellent sand handling capabilities.

Stainless Steel Klipring: Reuseable and provides greater ease of assembly and disassembly.

Hex Shaft Design: Six sided shaft eliminates impeller keys and provides more positive drive.

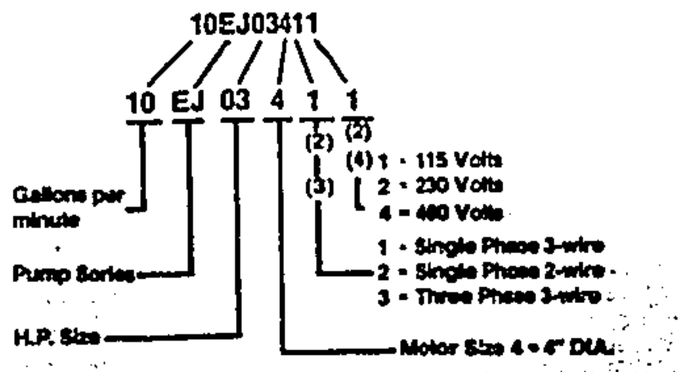
Stainless Steel Shaft Sleeve: Corrosion resistant.

Stainless Steel Shaft Coupling: Provides greater strength and corrosion resistance.

Franklin Electric Motor: 100% corrosion resistant stainless steel construction thru 1½ HP. Water filled design provides a constant supply of lubrication. Hermetically sealed stator assures moisture free windings. For added protection, a built-in surge arrester is provided on ½ HP thru 3 HP, single phase. Durable Kingsbury type thrust bearing absorbs all thrust. Replaceable motor lead assembly.

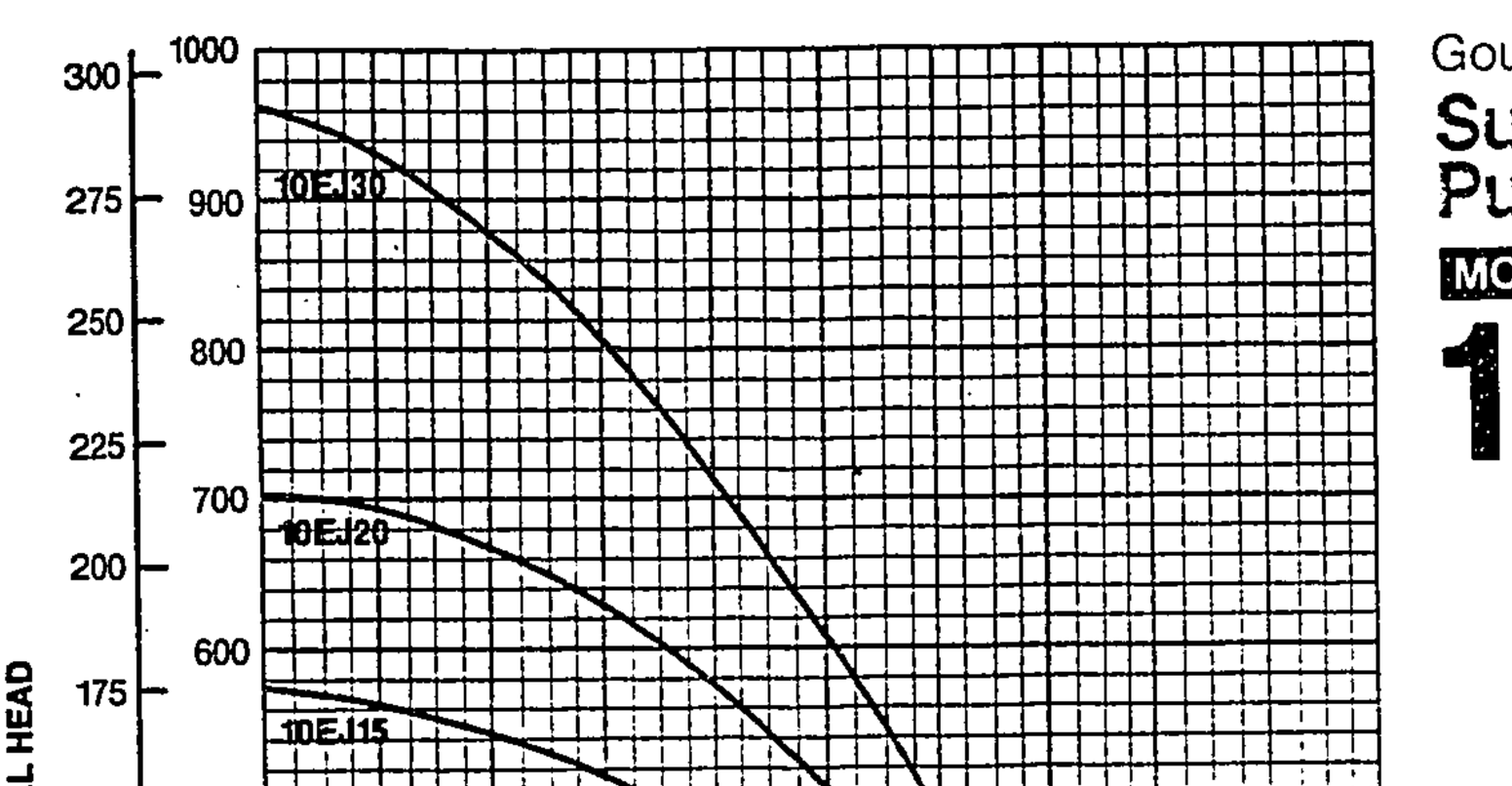
System Components: The basic pump-motor unit includes a three wire motor lead. A control box is supplied as standard equipment with all single phase units. A magnetic starter with three-leg protection is required with all three-phase units. Magnetic starter and heaters must be ordered separately.

NUMBER CODE



©1985 Goulds Pumps, Inc.

Effective July, 1985



Goulds

Submersible Pumps

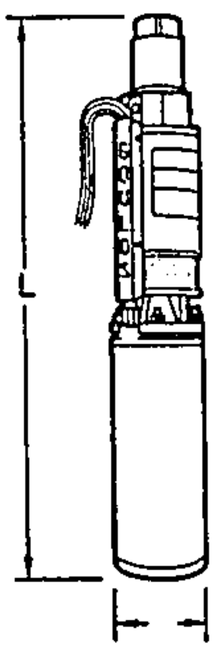
MODEL

10EJ

DIMENSIONS AND WEIGHTS

(All dimensions in inches and weights in lbs.) (Do not use for construction purposes.)

HP	Stages	3-Wire Model No.	3-Wire Length	2-Wire Model No.	2-Wire Length	Weight
⅓	5	10EJ03411, 12	21⅛	10EJ03421, 22	21⅛	28
½	7	10EJ05411, 12	23½	10EJ05421, 22	24	31
¾	10	10EJ07412	26¾	10EJ07422	27⅛	34
1	12	10EJ10412	29½	10EJ10422	30	39
1½	17	10EJ15412	36¼	10EJ15422	35½	50
1½	17	10EJ15432, 34	34¼	—	—	45
2	21	10EJ20412	45½	—	—	68
2	21	10EJ20432, 34	43	—	—	54
3	30	10EJ30412	55½	—	—	76
3	30	10EJ30432, 34	52⅛	—	—	63



DISCHARGE 1" NPT up to ¾ HP
1¼" NPT thru 3 HP

SENECA FALLS, NEW YORK 13148

SPECIFICATIONS ARE SUBJECT TO CHANGE WITHOUT NOTICE.

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DISINFECTION OF WATER RESERVOIRS

SECTION 201A

	Title	Page
201A.1	General	201A-1
201A.2	Liquid Chlorine Use	201A-1
201A.3	Sodium Hypochlorite Use	201A-1-2
201A.4	Method of Chlorination	201A-2
201A.5	Bacteriological Sampling and Testing	201A-2-3
201A.6	Tables	201A-4

201A.1 General. Sodium hypochlorite is available in liquid form in glass, rubber-lined, or plastic containers typically ranging in size from 1 qt. to 5 gal; containers of 30 gal. or larger size may be available in some areas. Sodium hypochlorite contains approximately from 5% to 15% available chlorine by volume, but care must be used in control of conditions and length of storage to minimize its deterioration. The Method of Chlorination shall be to chlorinate the full storage facility with water having a free chlorine residual of 2 mg/l after 24 hours.

201A.2 Liquid Chlorine Use. Liquid chlorine shall be introduced into the water filling the storage facility to give a uniform chlorine concentration during the entire filling operation. Portable chlorination equipment shall be carefully operated and shall include a liquid chlorine cylinder, gas-flow chlorinator, chlorine ejector, safety equipment, and an appropriate solution tube to inject the high concentration chlorine solution into the filling water. The solution tube shall be inserted through an appropriate valve located on the inlet pipe and near the storage facility such that the chlorine solution will mix readily with the inflowing water.

201A.3 Sodium Hypochlorite Use. Sodium hypochlorite shall be applied to the water entering the storage facility by means of a chemical feed pump, or shall be applied by hand-pouring into the storage facility and allowing the inflowing water to provide the desired mixing.

When a chemical feed pump is used, the concentrated chlorine solution shall be pumped through an appropriate solution tube to inject the high concentration chlorine solution at a rate which will give a uniform chlorine concentration in the filling water. The solution tube shall be inserted through an appropriate valve located on the inlet pipe and near the storage facility, or through an appropriate valve located on the storage facility such that the chlorine solution will mix readily with the filling water.

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When the sodium hypochlorite is poured into the storage facility, the filling of the storage facility shall begin immediately thereafter or as soon as any removed manhole covers can be closed. The sodium hypochlorite may be poured through the cleanout or inspection manhole in the lower course or level of the storage facility, or in the riser pipe of an elevated tank, or through the roof manhole. The sodium hypochlorite shall be poured into water in the storage facility when such water is not more than 3 feet in depth, or less than 1 foot in depth or as close thereto as manhole locations permit.

201A.4 Method of Chlorination. Water and chlorine shall be added to the storage facility in amounts such that initially the solution will contain 50mg/L available chlorine and will fill approximately 5% of the total storage volume, and this solution shall be held in the storage facility for a period of not less than 6 hours. The storage facility shall then be filled to the overflow level by flowing potable water into the highly chlorinated water, and shall be held full for a period of not less than 24 hours. All highly chlorinated water shall then be purged from the drain piping. Then, subject to satisfactory bacteriological testing and acceptable aesthetic quality, the remaining water may be delivered to the distribution system.

Chlorine shall be added to the storage facility by the method described in Section 201A.2 or 201A.3. The actual volume of the 50mg/L chlorine solution shall be such that after the solution is mixed with filling water and the storage facility is held full for 24 hours there will be free chlorine residual of not less than 2mg/L. See tables on page 201A-4.

201A.5 Bacteriological Sampling and Testing. After the chlorination procedure is completed, and before the storage facility is placed in service, water from the full facility shall be sampled and tested for coliform organisms in accordance with the latest edition of Standard Methods for the Examination of Water and Wastewater. The testing shall be by either the multiple tube fermentation technique or the membrane filter technique.

Such water also should be tested to assure that no offensive odor exists due to chlorine reactions or excess chlorine residual.

If the above sample is negative, then the storage facility may be placed in service. If the above sample shows the presence of coliform bacteria, then repeat samples shall be taken until two consecutive samples are negative, or the storage facility shall again be subjected to disinfection.

The samples shall be taken from a sample tap on the outlet piping from the storage facility, or from a sample tap connected directly to the storage facility. In either case, the operation shall be such as to assure that the sample collected is actually from water that has been in the storage facility.

During the disinfection operation and the required sampling of water from the storage facility, it is recommended that samples be taken from water inflowing to the storage facility to determine if coliforms are present in the typical potable water source.

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TABLE A.1

Amounts of Chemicals Required to Give Various Chlorine Concentrations in 100 000 gal of Water

Desired* Chlorine Concentration in Water mg/L	Pounds of Liquid Chlorine Required	Gallons of Sodium Hypochlorite Required			Pounds of Calcium Hypochlorite Required
		5% Available Chlorine	10% Available Chlorine	15% Available Chlorine	65% Available Chlorine
2	1.7	3.9	2.0	1.3	2.6
10	8.3	19.4	9.9	6.7	12.8
50	42.0	97.0	49.6	33.4	64.0

*Amounts of chemicals are for initial concentrations of available chlorine. Allowance may need to be made for chlorine depletion where low concentrations are held for extended time periods.

TABLE A.2

Amounts of Chemicals Required to Neutralize Various Residual Chlorine Concentrations in 100 000 gal of Water

Residual Chlorine Concentration mg/L	Pounds of Chemical Required			
	Sulfur Dioxide (SO_2)	Sodium Bisulfite ($NaHSO_3$)	Sodium Sulfite (Na_2SO_3)	Sodium Thiosulfate ($Na_2S_2O_3 \cdot 5H_2O$)
1	0.8	1.2	1.4	1.2
2	1.7	2.5	2.9	2.4
10	8.3	12.5	14.6	12.0
50	41.7	62.6	73.0	60.0

TO BE COMPLETED BY ACT 250
Act 250 jurisdiction: ___ YES X NO
This jurisdictional determination is based upon information provided and the following:
less than 10 acres involved land Env Bd Rule(2)(A)(4)
(Mitchell

THIS IS AN ADVISORY OPINION AND MAY BE APPEALED. ANY PARTY WHO DISAGREES WITH THIS OPINION MAY REQUEST A REVIEW DIRECTLY BY THE EXECUTIVE OFFICER OF THE ENVIRONMENTAL BOARD (ACT 250), OR THE DIRECTOR OF THE PROTECTION DIVISION, AS APPROPRIATE WITHIN 30 DAYS OF DECISION.

We also recommend that you contact the following Vermont State agencies because this project may need approval under their programs:

1. AGENCY OF NATURAL RESOURCES, Waterbury, Vt. 05676

_____ Air Pollution Control (244-8731)
_____ Hydro Electric Projects (244-7347)
_____ Dam Operations & Review (244-8755)
_____ Management of Lakes & Ponds (244-6951)
_____ Discharge of Temporary Pollution & Stormwater Permit (244-5674)
_____ Waste Management/Sanitary Landfills (244-8702)
_____ Industrial Furnace/Boiler Conversion/ Installation (244-8731)
_____ Wetlands (244-6951)
_____ Water Quality Certificate, if Federal Permit Required (244-6951)
_____ Use of Chemicals in State Waters (244-5638)
_____ Indirect Discharge (greater than 6,500 gpd) (244-5674)
_____ Bulk Fuel Chemical Storage/Hazardous Wastes (244-8702)
_____ Industrial Process Air Emissions (244-8731)
_____ Underground Storage Tanks (244-8702)

2. DEPARTMENT OF LABOR AND INDUSTRY, Montpelier, VT 05602 - Tel. 828-2106

_____ Electrical Wiring Approval
_____ Fire Prevention Approval
_____ Access for the Handicapped Approval
_____ Boilers and Pressure Vessels
_____ Plumbing in Public Buildings
_____ Tramways and Ski Facilities
_____ Storage of Flammable Liquids, Explosives
_____ Elevators
_____ Plumbing in Single Family Residences Served by Public Water/Sewer w/10 or More Customers

3. DEPARTMENT OF HEALTH, Burlington, VT 05401 - Tel. 863-7220 (Contact_____)

_____ Food, Lodging, Bakeries & Children's Camps
_____ Nursing Homes, Hospitals, & Homes for the Aged
_____ Protection of Public Water Source
_____ Public Water Systems
_____ Extensions of Water Lines over 500'

4. DISTRICT TRANSPORTATION ADMINISTRATOR (Consult telephone directory under VERMONT, STATE OF, Highways)

_____ Access to State Highways (Residential Driveways)
_____ Development within 500' of a Limited Access Highway

? (5.) AGENCY OF TRANSPORTATION, Montpelier, VT 05602 - Tel. 828-2823

_____ Signs (Travel Information Council)(828-2651)
_____ Airports and Landing Strips (828-2828)
_____ Junkyards (828-2587)
? _____ Construction within State Highway Right-of-Way (Utilities, Signs, Driveways) (828-2653)

6. DEPARTMENT OF AGRICULTURE, Montpelier, VT 05602 (Contact:_____)

_____ Use of Pesticides (828-2428)
_____ Milk Processing Facilities (828-2433)
_____ Primary Agricultural Soils (828-2504)
_____ Slaughter Houses (828-2426)
_____ Animal Shelters/Kennels (828-2421)

7. AGENCY OF HUMAN SERVICES, Waterbury, VT 05676 - Tel. 241-2158 (Contact:_____)

_____ Day Care Facilities
_____ Community Care Homes
_____ Residential Child Care Facilities

8. PUBLIC SERVICE BOARD (828-2358)/PUBLIC SERVICE DEPARTMENT (828-2811)

_____ Hydro Projects
_____ Sells Water to 2 or More Buildings.
_____ Alternative Sources of Energy
_____ Energy Conservation Measures

9. _____ DIVISION FOR HISTORIC PRESERVATION, Montpelier, VT 05602 - Tel. 828-3226

(10.) ✓ LOCAL PERMITS - Contact Your Town Officials

(11.) OTHER: Dept of Education

Prepared By Faye Clarke Department of Environmental Conservation

Copies TO: _____ / _____ / _____

State of Vermont

Department of Fish and Wildlife
Department of Forests, Parks and Recreation
Department of Environmental Conservation
State Geologist
Natural Resources Conservation Council

AGENCY OF NATURAL RESOURCES
Department of Environmental Conservation
Barre Regional Office
324 North Main Street
Barre, VT 05641

Telephone 479-3621

May 27, 1991

Waitsfield School District
Waitsfield, VT 05673

Dear Applicant:

RE: WW-5-0204-1 , replace water reservoir, WAITSFIELD

We received your completed application for the above referenced project on May 23, 1991, including a fee of $0 paid by check number . This application falls under the Water/Wastewater: Less than 500 GPD program; and under the Performance Standards for this program area, we have a maximum of 30 days of "in-house" time for our review of your application.

Based upon the present workload, we anticipate review of your project within the next one to two weeks.

The checklist on the back of this letter indicates other State agencies you should contact regarding any approvals you may need under their programs. The information contained in your application has been made available to an Act 250 District Coordinator for a determination of whether your project will also require an Act 250 permit. If you have questions relating to Act 250 jurisdiction, a District Coordinator may be reached at 479-3621 for Washington and Lamoille County projects and for projects in the towns of Orange, Washington, and Williamstown. For other Orange County projects call the District Coordinator at 886-2215.

If you have not already done so, you should check with town officials regarding any local requirements you may need to meet.

For the Division of Protection

Alida J. Conti
Administrative Secretary

cc: Act 250 District Coordinator
WAITSFIELD Planning Commission
Charles Grenier
Waitsfield Fire Department

SEE REVERSE FOR OTHER STATE AGENCIES THAT MAY REQUIRE APPROVAL OF THIS PROJECT